Marmik Shah

shahmarmik1@gmail.com – +91 87676 62635 – Mumbai, India – LinkedIn – GitHub – Portfolio

EXPERIENCE

Loylty Rewardz - A Bill Desk Group Company *Mumbai, India*
Senior Software Developer *Apr 2025 – Present*

- Architected and implemented a new channel loyalty product for manufacturers and finance companies, serving DSAs, agents, and channel partners across the distribution lifecycle with KPI management, leader-boards, tier computation, customer segmentation, and an approval module.
- Designed a real-time, API-driven microservices platform replacing T+1 bulk-file settlement, with 99% of transactions under 800ms; used Spring AOP for cross-cutting concerns and tracing, plus a webhook-based data-relay service.
- Built a low-latency balance service using caching, Bloom filters and database views, handling 300 TPS, letting partners fetch balances and burn points across corporates and LOBs in a single go.
- Built a unified vendor integration layer for multiple modules using strategy-based orchestration and Kong gateway, cutting configuration and development effort when onboarding clients.
- Designed a bucket-based reward matrix framework and optimized enrichment and tenant loading, reducing database calls and improving Hikari connection-pool use and startup time.
- Built admin panels and development tools using AI, so client, product and RM teams manage configurations independently; migrated services to Java 21 and handled SSL and TLS certificate migrations.

Software Developer *Jun 2023 – Mar 2025*

- Built a Kafka-driven ingestion platform processing 1M+ records daily, leveraging polling and web-hook patterns for asynchronous status tracking and reconciliation to catch missed status updates.
- Designed an enterprise-wide loyalty program letting one program span multiple lines of business, with generic APIs.
- Developed a milestone-based journey rewards product on GraphQL; architected a profile-driven multi-database setup and a strategy-driven post-award layer for cashback and other redemption modules.
- Designed a parallel-processing lock for distributed ad hoc transactions and implemented the budget module to control reward spend, with threshold-based actions and real-time budget updates.
- Achieved zero downtime on production services with keep-alive, virtual IPs and master-slave HA; cut Redis memory by over 50% with a purging policy, a cluster setup and tuned RDB snapshots.
- Deployed micro-services architecture on a cloud Kubernetes cluster; implemented observability using ELK, Prometheus, and Grafana.

ComED Learning *Apr 2022 – Jul 2022*
Software Development Intern *Remote*

- Developed UI/UX components and backend functionality for the ComED Learning platform.
- Integrated backend APIs for data exchange and dynamic content rendering; debugged application issues and performed functional testing to improve reliability.

KJ Somaiya College Of Engineering *Feb 2021 – Apr 2021*
Web Development Intern *Mumbai, India*

- Developed the website for the college cultural fest, Symphony, collabrating with cultural and design teams.
- Built responsive web interfaces with interactive components and animations, optimizing layouts for different screen sizes.

PROJECTS

Mumbai Police Web Portal — Built a full-stack complaint portal with station routing and status tracking (Django, PostgreSQL, Leaflet.js).

Waiterio Application — Developed a client-server desktop app where waiters place orders and the chef fulfills (Java, Swing, Sockets).

Newzer — A personalized news recommendation website that curates and summarizes news articles based on users' interests, preferences, and feedback using a recommendation algorithm. (Django, Javascript, ML Algorithms).

EDUCATION

KJ Somaiya College Of Engineering, *B.Tech in Computer Engineering | CGPI – 8.93* 2021–2023
Mumbai University, *SSC – 89.80% | HSC – 84.62%* 2017–2019

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, Python, SQL, C, C++, Solidity
Frameworks & Libraries: Spring, Spring Boot, Node.js, Django, React, Next.js, Vue.js, SASS, Bootstrap
Database & Caches: MySQL, PostgreSQL, Greenplum, SQL Server, MongoDB, Supabase, Redis, Neo4j
AI Tools: Claude, GPT, Cursor, Kiro, Copilot
Devops & Infra: Jenkins, Prometheus, Grafana, Newrelic, VueNet, ELK, Alertmanager, Linux, Bash, Nginx, Crontab, High Availability
Tools & Platforms: Docker, Kubernetes, Git, Azure, Apache Camel, Airflow, Kong Gateway, IPFS, Parcel
Messaging, API & Query Languages: Apache Kafka, Rabbit MQ, REST, JPA, GraphQL, Bloom Filters

ACHIEVEMENTS

Awards: Rising Star (2023–24), Extraordinary Diligence (2024–25), Think Out of the Box (2025–26).
Achievements: Won first place in HackerBash (CSI) hackathon as Team GrowMore. Project - Co-Vaccine Web portal